This is sample data provided for CALIBER purposes only	**CENTRIFUGAL PUMP DATA SHEET**	DOC NO: TJC-LLD-DS-GA-1201A REV: 3

ITEM / TAG NO.	GA-1201A	**EQUIPMENT ID**	EQ-1000012011
EQUIPMENT NAME	HEXANE FEED PUMP	**TYPE**	Centrifugal Pump (API 610 OH2)
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	1200 - FEED PREPARATION & PURIFICATION
FUNCTIONAL LOC.	TJC-LLD-1200-01	**CRITICALITY**	HIGH CRITICAL
SERVICE	Hexane feed transfer to 1st Polymerization Reactor DC-4501		

DESIGN & MECHANICAL DATA

PUMP TYPE	HORIZONTAL, END SUCTION, SINGLE STAGE	**CASING MATERIAL**	ASTM A351 CF8M (SS316)
MODEL	3x2x10 CDX-M	**IMPELLER MATERIAL**	ASTM A351 CF8M (SS316)
RATED FLOW	45 m3/h	**SHAFT MATERIAL**	AISI 4140
RATED HEAD	120 m	**MECHANICAL SEAL**	JOHN CRANE T2100, SINGLE
NPSH REQUIRED	3.5 m	**SEAL FLUSH PLAN**	API PLAN 11 + PLAN 62 QUENCH
DIFF. PRESSURE	8.6 bar	**BEARING TYPE**	DE: 7310 BECBM / NDE: 6310 C3
PUMPING TEMP.	40 degC	**LUBRICATION**	OIL BATH, ISO VG 68
SPECIFIC GRAVITY	0.66 (n-Hexane)	**COUPLING**	FLEXIBLE SPACER, REXNORD
VISCOSITY	0.31 cP		

DRIVER / MOTOR / HEATER DATA

OUTPUT	30 kW	**ENCLOSURE**	TEFC IP55
POLES	2 POLE	**AREA CLASSIFICATION**	ZONE 2, IIA T3
SPEED	2970 rpm	**EX PROTECTION**	Ex nA IIC T3 Gc
VOLTAGE	380 V / 3 Ph / 50 Hz	**FRAME**	IEC 200L
RATED CURRENT	56 A	**WEIGHT (MOTOR)**	295 kg
INSULATION CLASS	F (Temp rise B)		

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	Electric Motor, TEFC	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

1. Cross-reference for tag GA-1201A: GA Drawing, P&ID; TJC-LLD-PID-1201, Interlock SEQ-1201, OPL sheets and Maintenance History.
2. Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.

This is sample data provided for CALIBER purposes only	**ROTARY DRYER MOTOR DATA SHEET**	DOC NO: TJC-LLD-DS-YD-2301 REV: 3

SDK Polyolefin Expansion Project

ITEM / TAG NO.	YD-2301	**EQUIPMENT ID**	EQ-1000023011
EQUIPMENT NAME	POLYMER FLUID BED DRYER	**TYPE**	Rotary Steam-Tube Dryer
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	2300 - POLYMER SEPARATION & DRYING
FUNCTIONAL LOC.	TJC-LLD-2300-01	**CRITICALITY**	HIGH CRITICAL
SERVICE	Drying of LLDPE polymer powder from centrifuge to < 300 ppm moisture		

DESIGN & MECHANICAL DATA

DRYER TYPE	ROTARY STEAM-TUBE, INDIRECT	**OPERATING TEMP.**	95-110 degC
ITEM NO.	YD-2301	**BODY MATERIAL**	SM400B
DRUM DIAMETER	1800 mm	**ROTOR SHAFT**	SM400B / SS400
DRUM LENGTH	9000 mm	**GLAND PACKING**	PTFE (PILLER 4526L)
ROTATION SPEED	4 rpm	**OIL SEAL**	NBR
HEATING MEDIUM	LP STEAM 4.5 barg	**O-RING**	NBR
INERT MEDIUM	NITROGEN	**ROTARY JOINT**	ASSEMBLY, KADANT
DESIGN TEMP.	150 degC	**DRIVE**	CHAIN & SPROCKET SCM440

DRIVER / MOTOR / HEATER DATA

OUTPUT	22 kW	**ENCLOSURE**	TEFC IP55
POLES	4 POLE	**AREA CLASSIFICATION**	ZONE 2, IIA T3
SPEED	1450 rpm	**EX PROTECTION**	Ex nA IIC T3 Gc
VOLTAGE	380 V / 3 Ph / 50 Hz	**FRAME**	IEC 180L
RATED CURRENT	43 A	**WEIGHT (MOTOR+GEARBOX)**	833 kg
INSULATION CLASS	F		

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	Electric Motor via Chain-Sprocket, TEFC	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

1. Cross-reference for tag YD-2301: GA Drawing, P&ID; TJC-LLD-PID-2301, Interlock SEQ-5500, OPL sheets and Maintenance History.
2. Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.

This is sample data provided for CALIBER purposes only	**FIXED-BED REACTOR DATA SHEET**	DOC NO: TJC-LLD-DS-DC-3401A REV: 3

SDK Polyolefin Expansion Project

ITEM / TAG NO.	DC-3401A	**EQUIPMENT ID**	EQ-1000034011
EQUIPMENT NAME	CATALYST REDUCTION REACTOR	**TYPE**	Fixed-Bed Reactor (Pd Catalyst)
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	3400 - CATALYST PREPARATION
FUNCTIONAL LOC.	TJC-LLD-3400-01	**CRITICALITY**	HIGH CRITICAL
SERVICE	Reduction of PdO catalyst to Pd using H2 (PdO + H2 -> Pd)		

DESIGN & MECHANICAL DATA

REACTOR TYPE	VERTICAL FIXED-BED, DOWNFLOW	**CATALYST**	Pd / Al2O3 (PdO reduced to Pd)
ITEM NO.	DC-3401A	**CATALYST VOLUME**	3.2 m3
DIAMETER (ID)	1200 mm	**SHELL MATERIAL**	SA-516 Gr.70 + SS304 clad
TANGENT-TANGENT	4500 mm	**INTERNALS**	SS304 support grid + hold-down
DESIGN PRESSURE	6 barg / FV	**HEATER**	EA-3401 ELECTRIC PRE-HEATER 180 kW
OPERATING PRESSURE	2 kg/cm2g	**INERT MEDIUM**	NITROGEN
DESIGN TEMP.	250 degC	**REDUCTANT**	HYDROGEN (diluted in N2)
OPERATING TEMP.	180-210 degC	**PSV**	PSV-3401 set 6 barg

PERFORMANCE / NOZZLE DATA

EA-3401 HEATER OUTPUT	180 kW	**AREA CLASSIFICATION**	ZONE 1, IIC T3
HEATER VOLTAGE	380 V / 3 Ph / 50 Hz	**EX PROTECTION**	Ex d IIC T3
HEATER CURRENT	273 A	**TIC CONTROL**	TIC-17343 (heater outlet)
HEATER TYPE	IN-LINE ELECTRIC, N2 STREAM	**BED TC's**	TE-3401-1..8 multipoint

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	None (static vessel) - assoc. heater EA-3401	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

1. Cross-reference for tag DC-3401A: GA Drawing, P&ID TJC-LLD-PID-3401, Interlock SEQ-3401, OPL sheets and Maintenance History.

2. Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.

This is sample data provided for CALIBER purposes only	**RECIPROCATING COMPRESSOR DATA SHEET**	DOC NO: TJC-LLD-DS-KC-4501 REV: 3

SDK Polyolefin Expansion Project

ITEM / TAG NO.	KC-4501	**EQUIPMENT ID**	EQ-1000045011
EQUIPMENT NAME	RECYCLE GAS COMPRESSOR	**TYPE**	Reciprocating Compressor (2-stage)
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	4500 - GAS RECOVERY & COMPRESSION
FUNCTIONAL LOC.	TJC-LLD-4500-01	**CRITICALITY**	HIGH CRITICAL
SERVICE	Compress recycle ethylene/hexane vapour back to reactor loop		

DESIGN & MECHANICAL DATA

COMPRESSOR TYPE	RECIPROCATING, 2-STAGE, 2-THROW	**CYLINDER MATERIAL**	NODULAR CAST IRON
ITEM NO.	KC-4501	**PISTON RINGS**	PTFE FILLED
CAPACITY (SUCTION)	1250 Nm3/h	**PACKING**	PTFE RINGS, DISTANCE PIECE
SUCTION PRESSURE	1.2 barg	**VALVES**	PLATE TYPE, SS
DISCHARGE PRESSURE	12.5 barg	**LUBE SYSTEM**	FORCED FEED + FRAME SPLASH
SUCTION TEMP.	40 degC	**SEAL GAS**	NITROGEN BUFFER
DISCHARGE TEMP.	115 degC (max)	**COOLING**	JACKET WATER + INTERCOOLER EA-4502
GAS	ETHYLENE / HEXANE / H2 MIX	**VIBRATION MON.**	VT-4501 CROSSHEAD

DRIVER / MOTOR / HEATER DATA

OUTPUT	250 kW	**ENCLOSURE**	WPII
POLES	4 POLE	**AREA CLASSIFICATION**	ZONE 2, IIB T3
SPEED	1485 rpm	**EX PROTECTION**	Ex px IIB T3
VOLTAGE	6000 V / 3 Ph / 50 Hz	**FRAME**	IEC 355
RATED CURRENT	30 A	**WEIGHT (MOTOR)**	1980 kg
INSULATION CLASS	F		

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	Electric Motor, 4-pole	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

1. Cross-reference for tag KC-4501: GA Drawing, P&ID TJC-LLD-PID-4501, Interlock SEQ-4501, OPL sheets and Maintenance History.

2. Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.

This is sample data provided for CALIBER purposes only	**SHELL & TUBE HEAT EXCHANGER DATA SHEET**	DOC NO: TJC-LLD-DS-EA-5601 REV: 3

SDK Polyolefin Expansion Project

ITEM / TAG NO.	EA-5601	**EQUIPMENT ID**	EQ-1000056011
EQUIPMENT NAME	SOLVENT HEATER	**TYPE**	Shell & Tube Heat Exchanger (BEM)
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	5600 - SOLVENT RECOVERY SYSTEM
FUNCTIONAL LOC.	TJC-LLD-5600-01	**CRITICALITY**	LOW CRITICAL
SERVICE	Heat recovered hexane solvent using MP steam before recycle		

DESIGN & MECHANICAL DATA

TEMA TYPE	BEM, HORIZONTAL	**TUBE LENGTH**	6000 mm
ITEM NO.	EA-5601	**NO. OF TUBES**	246
DUTY	1.85 MW	**TUBE PASSES**	2
SHELL SIDE FLUID	MP STEAM 10 barg	**TUBE MATERIAL**	SS304L
TUBE SIDE FLUID	HEXANE SOLVENT	**SHELL MATERIAL**	SA-516 Gr.70
SHELL DESIGN P/T	13 barg / 200 degC	**TUBESHEET**	SA-516-70 + SS304 overlay
TUBE DESIGN P/T	16 barg / 200 degC	**BAFFLE**	SEGMENTAL 25% CUT
TUBE OD x BWG	19.05 mm x 14 BWG	**GASKET**	SPIRAL WOUND SS316/GRAPHITE

PERFORMANCE / NOZZLE DATA

HEAT TRANSFER AREA	88 m2	**STEAM CONSUMPTION**	3100 kg/h
CLEAN U	620 W/m2K	**SOLVENT INLET T**	45 degC
SERVICE U	480 W/m2K	**SOLVENT OUTLET T**	95 degC
FOULING (TUBE)	0.00035 m2K/W	**PRESSURE DROP (TUBE)**	0.35 bar

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	None (static)	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

1. Cross-reference for tag EA-5601: GA Drawing, P&ID TJC-LLD-PID-5601, Interlock N/A (control loop only), OPL sheets and Maintenance History.

2. Material of construction per WPN-MAT-001. Refer to applicable control and protection settings before operation.

This is sample data provided for CALIBER purposes only	**CONTROL VALVE DATA SHEET**	DOC NO: TJC-LLD-DS-LV-6701 REV: 3

SDK Polyolefin Expansion Project

ITEM / TAG NO.	LV-6701	**EQUIPMENT ID**	EQ-1000067011
EQUIPMENT NAME	SEPARATOR LEVEL CONTROL VALVE	**TYPE**	Globe Control Valve + Diaphragm Actuator
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	6700 - PRODUCT SEPARATION
FUNCTIONAL LOC.	TJC-LLD-6700-01	**CRITICALITY**	LOW CRITICAL
SERVICE	Level control of LP Separator FA-6710 bottoms to downstream		

DESIGN & MECHANICAL DATA

VALVE TYPE	GLOBE, CAGE-GUIDED, ED	**FAIL ACTION**	FAIL CLOSED (FC)
TAG / ITEM	LV-6701	**ACTUATOR**	FISHER 667 SIZE 45
BODY SIZE	3 inch (DN80)	**POSITIONER**	DVC6200 HART, 4-20 mA
BODY RATING	ASME 300#	**AIR SUPPLY**	1.4 barg INSTRUMENT AIR
BODY MATERIAL	WCC CARBON STEEL	**PACKING**	PTFE V-RING
TRIM MATERIAL	SS316 + STELLITE	**BENCH SET**	0.2-1.0 barg
TRIM CHARACTERISTIC	EQUAL PERCENTAGE	**HANDWHEEL**	TOP-MOUNTED
RATED CV	110	**SERVICE FLUID**	HEXANE + POLYMER BOTTOMS

DRIVER / MOTOR / HEATER DATA

CONTROLLER	LIC-6701 (on FA-6710 level)	**ACTION ON AIR FAIL**	VALVE CLOSES
INPUT SIGNAL	4-20 mA	**SOLENOID**	XV-6701 (ESD)
STROKE	38 mm	**LIMIT SWITCH**	ZSL/ZSH-6701
LEAKAGE CLASS	ANSI CLASS IV		

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	Pneumatic Spring-Diaphragm Actuator	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

1. Cross-reference for tag LV-6701: GA Drawing, P&ID TJC-LLD-PID-6701, Interlock SEQ-6701, OPL sheets and Maintenance History.

2. Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.

This is sample data provided for CALIBER purposes only	COOLING TOWER FAN & DRIVE DATA SHEET	DOC NO: TJC-LLD-DS-CT-7801 REV: 3

SDK Polyolefin Expansion Project

ITEM / TAG NO.	CT-7801	**EQUIPMENT ID**	EQ-1000078011
EQUIPMENT NAME	COOLING TOWER CELL FAN	**TYPE**	Induced Draft Cooling Tower Fan
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	7800 - COOLING WATER SYSTEM
FUNCTIONAL LOC.	TJC-LLD-7800-01	**CRITICALITY**	NON CRITICAL
SERVICE	Induced draft cooling of circulating water for LLDPE unit		

DESIGN & MECHANICAL DATA

TOWER TYPE	INDUCED DRAFT, COUNTERFLOW, CELL 1	**BLADE MATERIAL**	FRP (GRP)
ITEM NO.	CT-7801	**HUB MATERIAL**	CAST ALUMINIUM
FAN DIAMETER	7300 mm (24 ft)	**DRIVE SHAFT**	COMPOSITE FLOATING SHAFT
FAN TYPE	AXIAL, 6 FRP BLADES	**WATER FLOW (CELL)**	2200 m3/h
AIR FLOW	1,150,000 m3/h	**RANGE / APPROACH**	10 degC / 4 degC
FAN SPEED	128 rpm	**DRIFT ELIMINATOR**	PVC CELLULAR
GEARBOX RATIO	11.3 : 1 RIGHT ANGLE	**FILL**	PVC FILM FILL
GEARBOX OIL	ISO VG 220 (4.5 L)		

DRIVER / MOTOR / HEATER DATA

OUTPUT	55 kW	**ENCLOSURE**	TEFC IP56 (wet duty)
POLES	6 POLE	**AREA CLASSIFICATION**	NON-HAZARDOUS
SPEED	970 rpm	**MOUNTING**	IN-DUCT, WET
VOLTAGE	380 V / 3 Ph / 50 Hz	**FRAME**	IEC 250M
RATED CURRENT	104 A	**WEIGHT (MOTOR)**	410 kg
INSULATION CLASS	F		

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	Electric Motor via Gearbox, TEFC	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

2. Cross-reference for tag CT-7801: GA Drawing, P&ID TJC-LLD-PID-7801, Interlock SEQ-7801, OPL sheets and Maintenance History.
3. Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.

This is sample data provided for CALIBER purposes only	PRESSURE VESSEL DATA SHEET	DOC NO: TJC-LLD-DS-FA-8901 REV: 3

SDK Polyolefin Expansion Project

ITEM / TAG NO.	FA-8901	**EQUIPMENT ID**	EQ-1000089011
EQUIPMENT NAME	REFLUX ACCUMULATOR DRUM	**TYPE**	Horizontal Pressure Vessel
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	**AREA**	8900 - SOLVENT FRACTIONATION
FUNCTIONAL LOC.	TJC-LLD-8900-01	**CRITICALITY**	HIGH CRITICAL
SERVICE	Collect DA-8910 column overhead condensate; supply reflux & product		

DESIGN & MECHANICAL DATA

VESSEL TYPE	HORIZONTAL DRUM WITH BOOT	**SHELL MATERIAL**	SA-516 Gr.70
ITEM NO.	FA-8901	**HEAD TYPE**	2:1 ELLIPSOIDAL
DIAMETER (ID)	1600 mm	**CORROSION ALLOW.**	3.0 mm
TANGENT-TANGENT	4800 mm	**BOOT**	WATER DRAW-OFF BOOT DN300
DESIGN PRESSURE	10 barg / FV	**PWHT**	YES
OPERATING PRESSURE	3.5 barg	**RADIOGRAPHY**	FULL (RT-1)
DESIGN TEMP.	120 degC	**PSV**	PSV-8901 set 10 barg
OPERATING TEMP.	68 degC	**N2 BLANKET**	PCV-8905 SPLIT-RANGE

PERFORMANCE / NOZZLE DATA

VOLUME	9.6 m3	**PRODUCT NOZZLE**	N3 4 inch
REFLUX PUMP	GA-8920 A/B	**BOOT DRAIN**	N4 2 inch
INLET NOZZLE	N1 8 inch (column overhead)	**PSV NOZZLE**	N5 3 inch
REFLUX NOZZLE	N2 6 inch	**MANWAY**	24 inch

VENDOR / MANUFACTURER	Vendor ABC	**DRIVER SUPPLIER**	Supplier XYZ
DRIVER TYPE	None (static)	**DATASHEET REV**	Rev 3 - ISSUED FOR OPERATION

NOTES:

1. Cross-reference for tag FA-8901: GA Drawing, P&ID; TJC-LLD-PID-8901, Interlock SEQ-8901, OPL sheets and Maintenance History.
2. Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.